The UK University Integration Bee
Round 3 Information

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Congratulations on making it to Round 3! In this document you will find details on Round 3 of the competition.

Structure of Round 3

Round 3 consists of three parts, as follows.

- Chainlink Round

  Mechanically based on *Who Wants to Be a Millionaire?*.

- Dichotomy Round

  Get two problems, choose one to solve. Catch is, the ones who choose and the one who solves are different people.

- Tournament Round

  Bracket tournament, but modified for the UK University Integration Bee format.

Further details of each round are in the following pages. As these rounds are newly-designed, we don't have examples on the website. Please read this document carefully and ensure you understand how each round works. The goal is to obtain as many points as you can from each of these rounds, and your team's final score is the sum of the points obtained in each round.

Schedule

Please refer to the following table for the schedule of Round 3.

Start	**End**	**Activity**
Before	1215	Arrival
1215	1220	Opening
1220	1230	Chainlink Round briefing
1230	1430	Chainlink Round (est. 120 mins)
1430	1445	Break
1445	1450	Dichotomy Round briefing
1450	1610	Dichotomy Round (est. 80 mins)
1610	1625	Break
1625	1630	Tournament Round briefing
1630	1850	Tournament Round (est. 140 mins)
1850	1900	Break
1900	After	Closing

Chainlink Round: 120 minutes

Maximum score: 150 points.

Organisation and flow of the round

In this round, there are 5 problem sets, each with 8 problems arranged in increasing order of difficulty (as judged by the organisers), for a total of 40 problems. Problems are given one at a time, just like the relay round in Round 2. Returning to a previous problem is not possible. Each problem has a limit of three attempts, skipping is restricted, and scoring is performed in terms of 'chains' of problems solved successfully. This round proceeds in two halves.

First Half

This half is loosely based on mechanics of the UK gameshows *The Weakest Link* and *Who Wants to Be a Millionaire?*. The idea is to solve consecutive problems correctly, starting from an easy problem and progressively getting harder, and the score is based on how many problems are solved, and whether the team ends the chain of solves on their own volition or fails a problem.

- This half lasts for five 'games'. Each game uses one of the problem sets (8 problems).
- Teams are given problems one at a time, starting from the easiest problem and increasing in difficulty.
- After each problem is solved successfully, a choice is given to the team: They can *bank* the chain of solves, or they can *continue* with the next problem.

  If the team chooses to bank, the game ends with Game End (Bank) status and the next game begins.
- If the team fails a problem, the game ends with Game End (Failed) status and the next game begins.
- Skips are not allowed (except for lifelines, see below).

Scoring is dependent on whether the Game End (Bank) or Game End (Failed) is achieved. The basic scoring is the following table.

Solved problems	0	1	2	3	4	5	6	7	8
Points	0	2	4	7	11	15	20	25	30

- If Game End (Bank) is achieved, the basic scoring is used as-is.
- If Game End (Failed) is achieved, the score is the highest safety net less than or equal to the number of solved problems. The default safety nets are $1, 3, 6$ solved problems.

Each team can use lifelines to assist them in this half. Multiple lifelines can be used in the same game.

- *Lifeline: Stop Loss*

  The team chooses a value of $\{2, 4, 5, 7\} \ni k$ solved problems to act as a safety net in addition to the default safety nets ($1, 3, 6$ solved problems). Applies only for the current game and does not carry over to future games. Must be used before the current game ends.

  Can be used a total of two times in this half.
- *Lifeline: Take the Money and Run*

  Voids the current problem (does not count as a solve) and achieves Game End (Bank). Must be used before the third attempt of the current problem.

  Essentially, this is a bank that can be used in the middle of a problem.

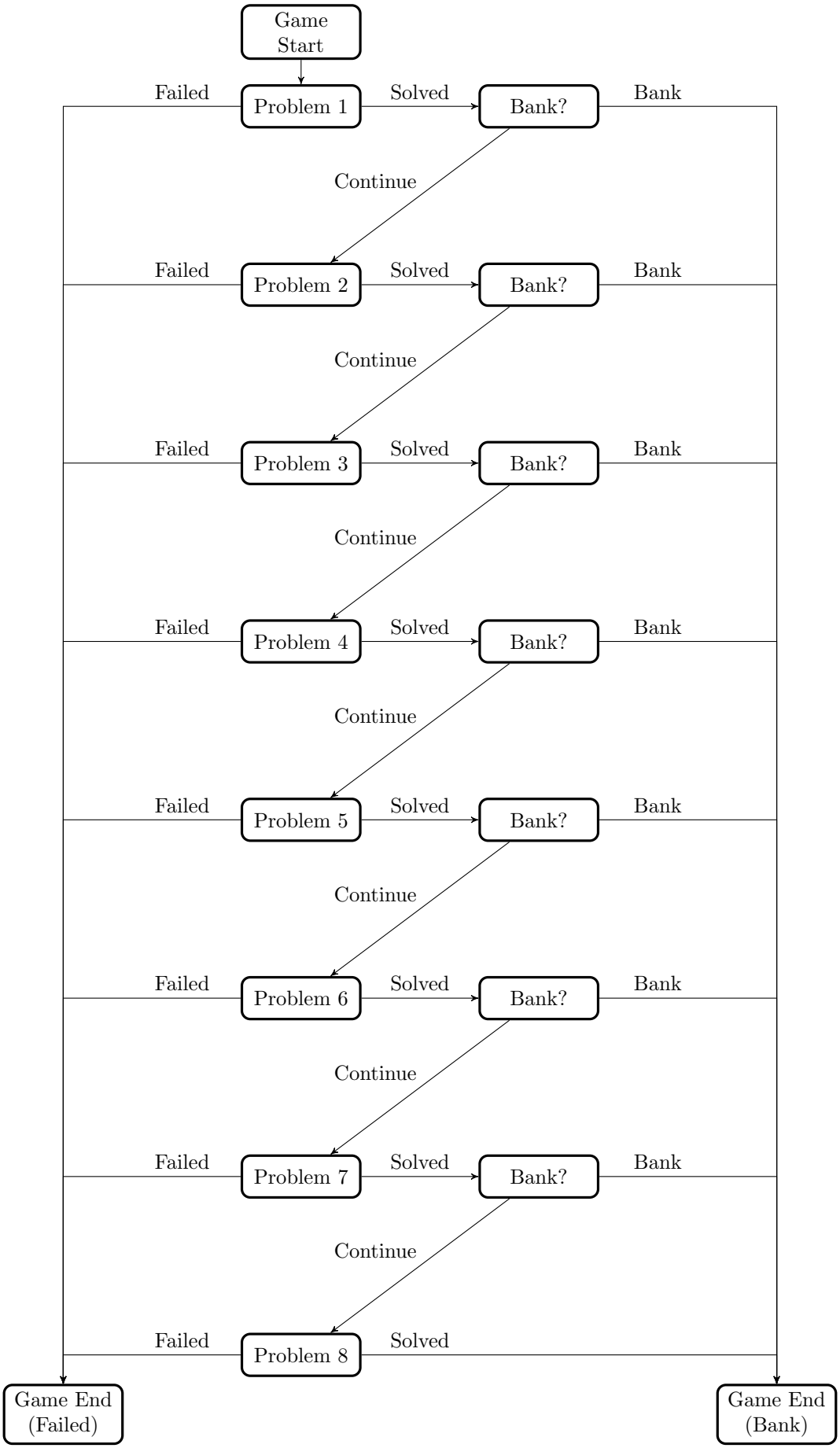
  Can be used a total of one time in this half.
- *Lifeline: This Problem is Dread*

  Voids the current problem (does not count as a solve) and moves on to the next problem immediately. If the current problem is the final problem of the current game, achieves Game End (Bank) instead. Must be used before the third attempt of the current problem.

  Essentially, this is a continue that can be used in the middle of a problem (no choice to bank given).

  Can be used a total of two times in this half.

The following illustrates the basic flow of each game.



Second Half

The second half consists of all unused problems in the first half.

- Problems are given to teams one at a time, starting from the unused problems in the first set (in difficulty order) and ending with the unused problems in the fifth set (in difficulty order).
- Like before, consecutive solves are advantageous. There is no penalty for failing a problem; the chain of solves just ends there and the next one begins.
- Skips are allowed. It counts as failing a problem.

Each team can use lifelines to assist them in this half.

- *Lifeline: This Problem is Dread*

  Voids the current problem (does not count as a solve) and moves on to the next problem without breaking the current chain of solves. Must be used before the third attempt of the current problem.

  Can be used a total of three times in this half.

Answer submission and scoring

The total number of problems to attempt is 40. Whatever is not attempted in the first half will appear in the second half. It is recommended that you obtain as many points as you can in the first half.

First Half

Submit your answers to your assigned grader. Three attempts allowed per problem. Skips are not allowed unless a lifeline is used. Only final answers will be checked. Scoring for each game follows the following table and depends on whether Game End (Bank) or Game End (Failed) is achieved.

Solved problems	0	1	2	3	4	5	6	7	8
Points	0	2	4	7	11	15	20	25	30

- If Game End (Bank) is achieved, the above scoring table is used as-is.
- If Game End (Failed) is achieved, the score is the highest safety net less than or equal to the number of solved problems. The default safety nets are $1, 3, 6$ solved problems.

  Example 1: The team solves 3 problems and fails the next. They get 7 points (safety net 3 solved problems).

  Example 2: The team solves 5 problems and fails the next. They get 7 points (safety net 3 solved problems).

  Example 3: The team solves 6 problems and fails the next. They get 20 points (safety net 6 solved problems).

  Using *Lifeline: Stop Loss* creates an additional safety net specified by the team. Note that the safety nets are based on number of solved problems (as opposed to problem number), thus *Lifeline: This Problem is Dread* does not change the required number of solves of the safety nets.

The score for this half is the sum of scores for each game.

Second Half

Submit your answers to your assigned grader. Three attempts allowed per problem. Skips are allowed (the problem is treated as failed/unsolved). Only final answers will be checked. The score for k consecutive solves in this segment (for $k \geq 1$) is $2k + \left\lfloor \frac{k-1}{2} \right\rfloor$. The score for this half is the sum of scores for each chain of consecutive solves. Using *Lifeline: This Problem is Dread* to void a problem does not break the chain, but neither does it count as a solved problem. For instance, if a team achieves

Solved?	Yes	Yes	Void	Yes	Yes	No	Yes	No	Yes	Yes

then their score for this half is $\left[2k + \left\lfloor \frac{k-1}{2} \right\rfloor\right]_{k=4} + \left[2k + \left\lfloor \frac{k-1}{2} \right\rfloor\right]_{k=1} + \left[2k + \left\lfloor \frac{k-1}{2} \right\rfloor\right]_{k=2}$.

Dichotomy Round: 80 minutes

Maximum score: 140 points.

Organisation and flow of the round

In this round, the team members are labeled A, B, C (if you have 3 people) or A, B (if you have 2 people). The round consists of three segments, each lasting 25 minutes and consisting of 14 problems arranged into 7 pairs. Only one problem in each pair will be submitted for scoring. If you are a 3-person team, in each segment two members will be the *selectors* and one member will be the *solver*.

- First segment: A, B are the selectors, C is the solver.
- Second segment: B, C are the selectors, A is the solver.
- Third segment: C, A are the selectors, B is the solver.

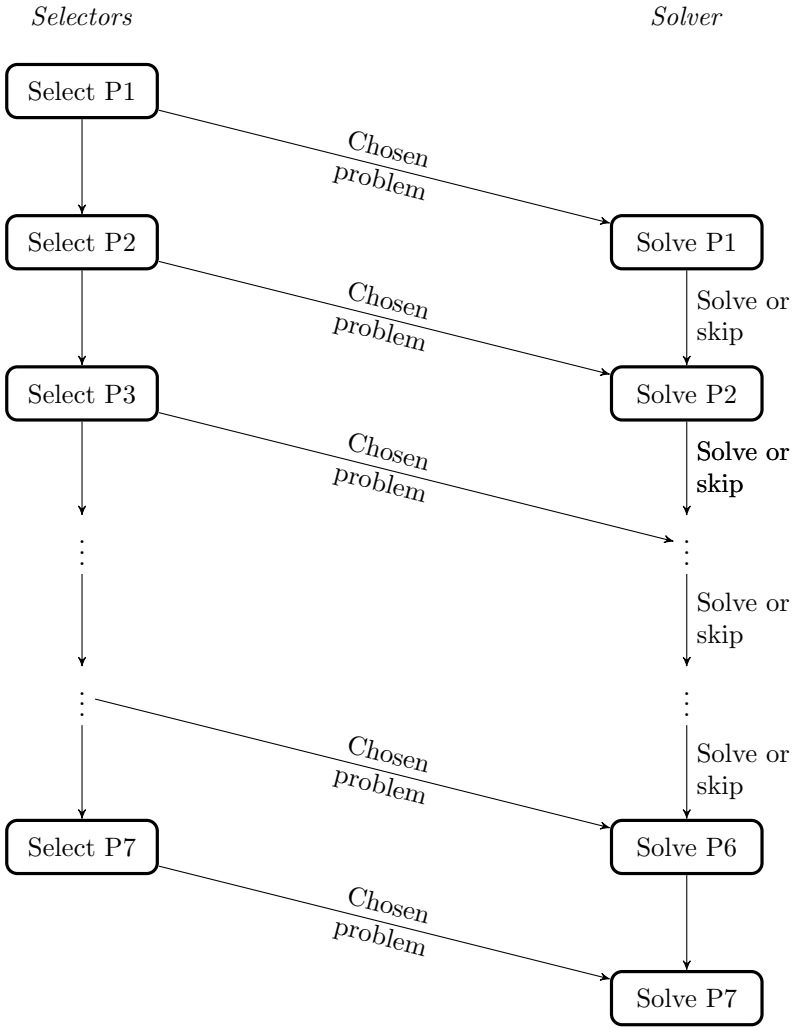
If you are a 2-person team, in each segment one member will be the *selector* and one member will be the *solver*.

- First segment: A is the selector, B is the solver.
- Second segment: B is the selector, A is the solver.
- Third segment: Up to you to choose the selector and the solver.

Selectors cannot communicate with the solver and vice-versa. Each pair of problems works as follows.

- The selectors will receive both problems in the pair. They must choose one of them for the solver to solve.
- The solver will then solve the chosen problem and submit their answer for scoring.

Selecting and solving run concurrently. While the selectors are choosing Problem $n + 1$, the solver is attempting Problem n. The following illustrates the workflow of each 25-minute segment.



The following gives more details of jobs of the selectors and the solver.

Selectors

You will receive a choice of two problems. Problems in each pair may be of uneven difficulty or require very different techniques, or they might not (we won't tell you). Your task is to choose one of these problems for your solver to attempt. You can't change your choice after your chosen problem is passed to the solver. You get the next pair of problems after your chosen problem in the current pair is passed to the solver.

Solver

Functionally, your job is like the Relay Round in Round 2, except that you're doing it alone and the problems you attempt are chosen by your team's selectors. Problems are given sequentially, one at a time. If you can't solve it you can skip it. You can't return to a previous problem. The next problem will only be given after you solve or skip the current problem, and after your selectors have chosen a problem for you.

Answer submission and scoring

Only the solver earns points, but it's the job of the selectors to make this as easy as possible for the solver. Submit your answers to your assigned grader. Unlimited attempts and skips are allowed, but a 10-second delay between each successive attempt for the same problem will be enforced to prevent rapid-fire random guessing. Only final answers will be checked. Scoring for each problem is as follows.

- Correct answer on the first attempt: 6 points
- Correct answer on the second or any later attempt: 2 points
- Skip: 0 points

In total there will be up to 21 problems attempted for a maximum base score of 126. Bonuses are applied, like in the Relay Round of Round 2. Solving 10 problems earns you a β. Solving 15 problems replaces your β with an α.

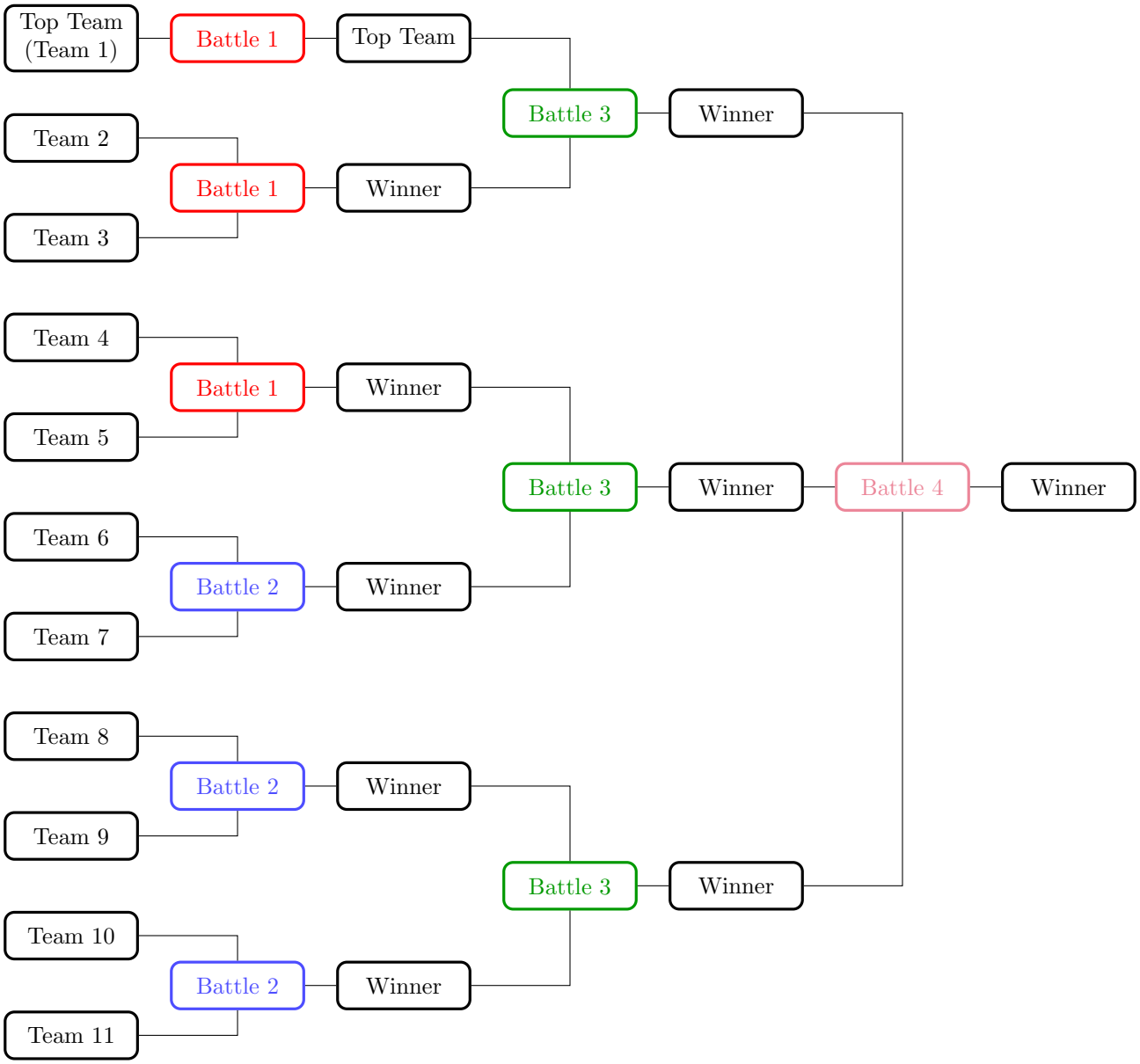
- β: Bonus 8 points.
- α: Bonus 14 points.

Tournament Round: 140 minutes

Maximum score: 169 points.

Organisation and flow of the round

This round is based on the head-to-head MIT style integration bee tournament battles, reworked to account for the presence of 11 teams, and the fact that this is a team-based competition with many components.



There are four battles in total. As can be seen, Battles 1 to 3 inclusive each take place across multiple pairs of teams. Think of it as a standard tournament face off between a pair of teams, except multiple such pairs face off against each other at the same with the same problems. In Battle 4, it's a face off between the final three surviving teams.

The Top Team refers to the team with the highest combined score of the Chainlink and Dichotomy Rounds. In their case, they will participate in Battle 1, but as a reward for good performance, they are guaranteed a progression to Battle 3. The organisers reserve the right to decide on seeding for Teams 2 to 11 (i.e they are not necessarily in ranking order).

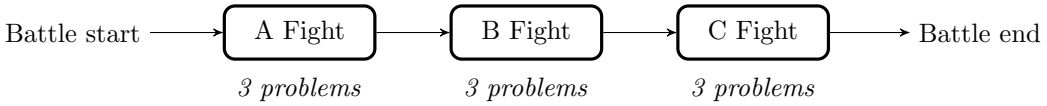
Battle format

As mentioned above, for Battles 1 to 3, think of it as multiple pairs facing off simultaneously on whiteboards, using the same problems. In Battle 4, think of it as three teams facing off against each other.

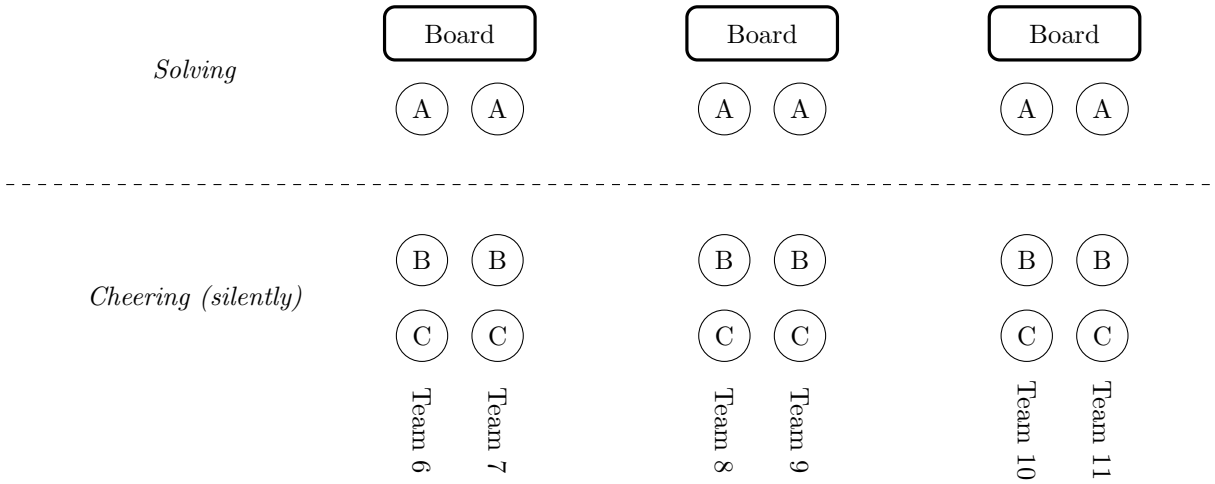
Each battle consists of three fights. Teams label their members A, B, C.

- A fight: A members.
- B fight: B members.
- C fight: C members (a two-person team can choose either member to participate).

Each fight consists of three problems, for a total of nine problems per battle. Two attempts per problem are allowed (unless Score Gamble is used, see below). Circle/box your answer to submit; if you don't, we will not accept it and you'll have wasted an attempt. The timeline of a battle is illustrated below.



As an example, the A fight in Battle 2 involving three pairs of teams is physically illustrated below. B fight and C fight are similar.



Battle parameters

- Battle 1:
  - 2.5 minutes per problem.
  - Two pairs of teams and the top team (5 teams total) will battle simultaneously.
- Battle 2:
  - 2.5 minutes per problem.
  - Three pairs of teams (6 teams total) will battle simultaneously.
- Battle 3:
  - 3 minutes per problem.
  - Three pairs of teams (6 teams total) will battle simultaneously.
- Battle 4:
  - 4 minutes per problem.
  - Three teams (3 teams total) will battle simultaneously.

Scoring and advancement

Teams are rewarded based on both ability (can you solve it) and speed (can you solve it faster). Generally, ability gets you points, and speed gets you to the next battle to try for even more points.

- Solving a problem within the allotted number of attempts gives the team the points for getting it right (see below for specific point values). This applies even if your rival has already solved the problem, so don't give up (to prevent cheating and to save time, we recommend you erase your answer once we verify it is right).
- In each competing pair (triple in the case of Battle 4), solving a problem before your rival gives you 1 BP[1].
- The winner of each fight is the contestant with the highest number of BP. The fight is a draw if BP are equal.

The overall winner of each pair (triple in the case of Battle 4) is decided by the following priority:

1. Number of fights won by the team.
2. Number of BP won by the team.
3. Number of solves by the team.

If the winner is still undecided, we proceed to a sudden death tiebreaker problem.

Other mechanics

There is a pair of mechanics to help teams raise their scores.

Score Gamble: Double or Nothing

This is double-or-nothing mechanic that can be invoked in battle by a contestant before a fight begins. It modifies the answering and scoring rules for that contestant in their fight.

- Score Gamble applies only to the person(s) who invoke(s) it.
- This is a straight double-or-nothing mechanic. The invoker has only one attempt for each problem in their fight. In exchange, the problems in their fight are all worth double points[2].

How to use:

- Before a fight begins, we will ask if any contestant wants to invoke Score Gamble. Contestants in the fight who wish to invoke Score Gamble should declare it at this time.
- Each team may invoke Score Gamble once per battle (i.e. for one fight per battle). You may of course decide not to use Score Gamble. It is up to you.

Redemption

Redemption is a mechanic applying to eliminated teams during Battles 3 and 4:

- In Battle 3, this mechanic involves the five eliminated teams.
- In Battle 4, this mechanic involves the eight eliminated teams.

It enables eliminated teams to try for more points in Battles 3 and 4. How it works:

- Eliminated teams may attempt the problems at their tables on paper for some extra points. The whole team can try together, and they can watch what the contestants at the boards are doing. If the team can solve the problem correctly *before any of the contestants at the boards manage it*, they get a few points.
- One attempt per team per problem.
- Submissions can only take place while no answer on the boards is being checked. Write it down and submit to the grader. Don't shout it out, the contestants are still solving at the boards!
- Does not apply to tiebreakers. Score Gamble does not apply in Redemption.

[1] BP is only used for deciding the winner of each fight/battle and does not convert to normal points. It is a local variable – it is reset after each battle.

[2] Does not affect BP. Solving the problem first still grants only 1 BP.

Answer submission and scoring

Scores depend on the battle, how many problems the team solves, uses of Score Gamble, as well as their final position in the tournament. In each battle, each team will attempt 9 problems in total. For battles, write your answer on the board and circle or box it up. Not circling/boxing it will lead to an automatic wasted attempt. Two attempts per problem, unless Score Gamble is used, then it's one attempt per problem.

Battles 1 and 2

Total available: 104 points (80 normal points + 24 points from Score Gamble).

- Correct answer: 8 points (16 points with Score Gamble)
- Winning the battle[†]: 8 points

[†]In the case of the Top Team, BP does not apply in Battle 1. Instead, they get these points if they solve more problems than the average (arithmetic mean) of the two winners of Battle 1.

Battle 3

Total available: 39 points (30 normal points + 9 points from Score Gamble).

- Correct answer: 3 points (6 points with Score Gamble)
- Winning the battle: 3 points

Battle 4

Total available: 26 points (20 normal points + 6 points from Score Gamble).

- Correct answer: 2 points (4 points with Score Gamble)
- Winning the battle: 2 points

Redemption

Total available: 27 points if eliminated in Battles 1 or 2. 9 points if eliminated in Battle 3.

Applies only to teams which have been eliminated from the tournament. Write your answers (don't shout it out!) on paper and submit to the grader. One attempt per team per problem is permitted and it must be made before anyone at the boards finish correctly. Submissions cannot be made while an answer on the board is being checked.

- Correct answer (Battle 3): 2 points
- Correct answer (Battle 4): 1 point